

LAPORAN DOKUMENTASI SOFTWARE QUALITY ASSURANCE (SQA)



Anggota kelompok:

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**PROGRAM STUDI TEKNIK INFORMATIKA
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BAB I – PENDAHULUAN

1.1 Latar Belakang

SayurSehat merupakan platform e-commerce berbasis RESTful API yang dirancang untuk mendukung rantai pasok sayuran segar dari produsen kepada konsumen. Sistem ini dibangun menggunakan Golang sebagai backend, PostgreSQL sebagai basis data, React.js sebagai antarmuka pengguna, dan Midtrans sebagai payment gateway.

Sebagai bagian dari proses pengembangan perangkat lunak yang berkualitas, dilakukan kegiatan Software Quality Assurance (SQA) yang menyeluruh menggunakan tiga pendekatan: White-box Testing, Black-box Testing, dan Grey-box Testing.

1.2 Tujuan Dokumentasi

1. Mendokumentasikan seluruh aktivitas pengujian perangkat lunak SayurSehat secara terstruktur.
2. Menyediakan bukti pengujian dari tiga metode (White-box, Black-box, Grey-box).
3. Melaporkan hasil temuan bug, perbaikan (fix), dan status akhir pengujian.
4. Menjadi referensi audit kualitas perangkat lunak untuk keperluan akademik MPPL.

1.3 Ringkasan Metode & Teknik Pengujian

Metode	PR GitHub	Teknik yang Digunakan	Modul Diuji
White-box Testing	#7, #8, #9, #10	Basic Path Testing, Code Walkthrough, Data Flow Testing, Formal Inspection, Loop Testing	Checkout (UC-02), Order, Payment
Black-box Testing	#16	Boundary Value Analysis, Equivalence Class Partitioning, Endurance, Performance, Sample Test	Checkout Module (UC-02)
Grey-box Testing	#15, #17	Matrix Testing, Orthogonal Array Testing, Regression Testing, Pattern Testing	Order Payment & Refund Status

1.4 Teknologi & Stack

Komponen	Teknologi	Keterangan
Backend	Golang	RESTful API
Database	PostgreSQL	Relational DB, ACID compliance
Frontend	React.js + Vite	Single Page Application
Payment	Midtrans	Payment Gateway
Version Control	GitHub	Repository, PR, Issue Tracking

BAB II – MANAJEMEN REPOSITORY GITHUB

2.1 Struktur Folder Dokumentasi

Repository proyek SayurSehat dikelola di GitHub dengan struktur folder dokumentasi pengujian sebagai berikut (dibuat melalui Issue #5 dan PR #6):

Path Folder	Isi	Keterangan
dokumentasi/whitebox/bukti	Bukti whitebox testing	Basic Path, Code Walkthrough, Data Flow, Formal Inspection, Loop Testing
dokumentasi/blackbox	Dokumen black-box testing	BVA, ECP, Endurance, Performance, Sample Test
dokumentasi/greybox/bukti	Bukti grey-box testing	Matrix, Orthogonal Array, Regression, Pattern Testing
dokumentasi/laporan-akhir/bukti	Laporan akhir & bukti	Laporan final SQA beserta semua bukti



2.2 Riwayat Pull Request

Seluruh perubahan kode dan dokumentasi dikelola melalui Pull Request (PR) dengan proses review sebelum di-merge ke branch main. Total terdapat 12 PR yang berhasil diselesaikan:

No	PR #	Judul PR	Status
1	#17	Menambahkan Dokumen pengujian Metode Grey Box	Merged ✓
2	#16	docs: add blackbox testing documentation	Merged ✓
3	#15	test: add regression suite for order payment and refund status transition	Merged ✓

4	#14	fix: error canceled order	Merged ✓
5	#13	Fix UI sidebar, perbaiki alur order, dan tambah skenario testing	Merged ✓
6	#12	fix: correct payment and order delivery workflow	Merged ✓
7	#11	fix: replace Vite favicon with SayurSehat icon	Merged ✓
8	#10	docs: add additional whitebox testing evidence	Merged ✓
9	#9	docs: add Basic Path checkout testing evidence	Merged ✓
10	#8	docs: add basic path checkout source code evidence	Merged ✓
11	#7	Menambahkan Dokumen Pengujian Metode Whitebox	Merged ✓
12	#6	Membuat struktur folder dokumentasi pengujian	Merged ✓

0 Open ✓ 15 Closed		Author	Label	Projects	Milestones	Reviews	Assignee	Sort
17	Menambahkan Dokumen pengujian Metode Grey Box ✓ documentation	irvansalfiansyah	Collaborator					1
16	docs: add blackbox testing documentation ✓	Nuryanfa	Owner					2
15	test: add regression suite for order payment and refund status transi... ✓	Nuryanfa	Owner					
14	fix: error canceled order ✗	Nuryanfa	Owner					
13	Fix UI sidebar, perbaiki alur order, dan tambah skenario testing ✓	Nuryanfa	Owner					
12	fix: correct payment and order delivery workflow ✓	Nuryanfa	Owner					
11	fix: replace Vite favicon with SayurSehat icon ✓	Nuryanfa	Owner					
10	docs: add additional whitebox testing evidence ✓	Nuryanfa	Owner					
9	docs: add Basic Path checkout testing evidence ✓	Nuryanfa	Owner					1
8	docs: add basic path checkout source code evidence ✓	Nuryanfa	Owner					2
7	Menambahkan Dokumen Pengujian Metode Whitebox ✓	irvansalfiansyah	Collaborator					3
6	Membuat struktur folder dokumentasi pengujian ✓	Nuryanfa	Owner					3

2.3 Issue Tracking

Manajemen tugas dilakukan melalui GitHub Issues. Issue #5 'Buatkan Folder Pengujian' menjadi dasar pembentukan struktur dokumentasi SQA, diselesaikan oleh Nuryanfa melalui PR #6 dengan membuat 5 folder: White-box Testing, Black-box Testing, Grey-box Testing, folder bukti per metode, dan folder laporan akhir.

Buatkan Folder Pengujian #5

 Closed  #6

**irvansalfiansyah** opened 2 weeks ago Collaborator 

Buatkan Folder White Box, Black Box dan Grey Box

Create sub-issue 

**Nuryanfa** 2 weeks ago Owner 

Struktur folder yang dibuat sebagai berikut

- folder White-box Testing.
- folder Black-box Testing.
- folder Grey-box Testing.
- folder bukti pada setiap metode.
- folder laporan akhir.



 **Nuryanfa** mentioned this 2 weeks ago

 Membuat struktur folder dokumentasi pengujian #6

 **Nuryanfa** closed this as completed in #6 2 weeks ago

BAB III – WHITE-BOX TESTING

3.1 Deskripsi Metode

White-box Testing dilakukan dengan menganalisis struktur internal kode Golang pada modul Checkout (UC-02). Dokumen pengujian ditambahkan melalui PR #7 oleh irvansalfiansyah dalam format Markdown (.md) dan PDF, mencakup lima teknik pengujian sebagaimana dijelaskan dalam deskripsi PR.

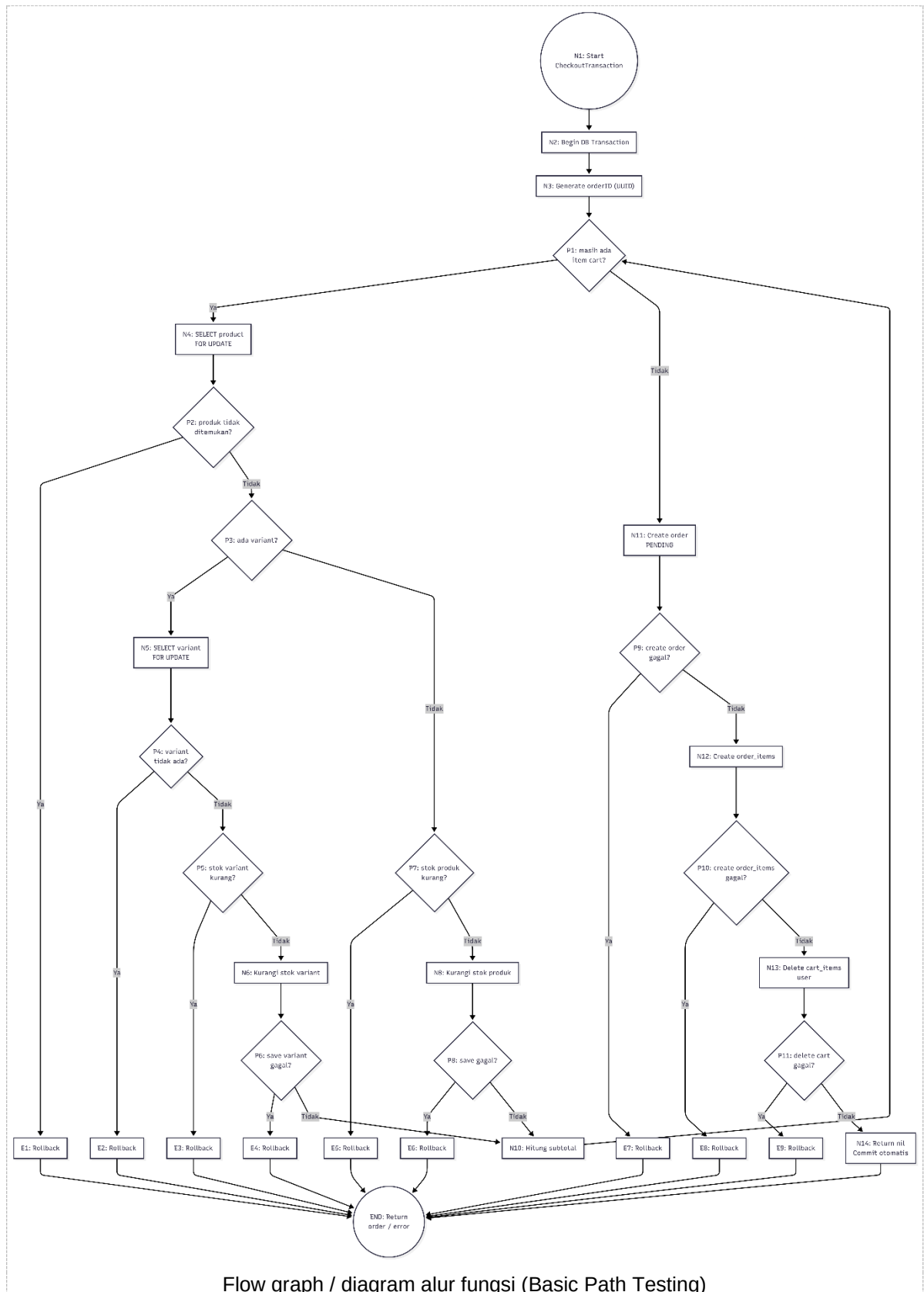
3.2 Teknik Pengujian White-box

No	Teknik	Deskripsi	Hasil / Keterangan
1	Basic Path Testing	Menentukan jalur independen berdasarkan flow graph dan cyclomatic complexity (CC) setiap fungsi	PASS
2	Code Walkthrough	Penelusuran kode secara manual untuk menemukan kesalahan logika sebelum dieksekusi	PASS
3	Data Flow Testing	Melacak aliran data dari definisi variabel hingga penggunaan untuk mendeteksi anomali	PASS
4	Formal Inspection	Review kode formal menggunakan checklist kriteria kualitas perangkat lunak	PASS
5	Loop Testing	Pengujian khusus pada struktur perulangan (simple, nested, concatenated loops)	PASS

3.3 Bukti Screenshot – White-box Testing

Bukti pengujian disimpan di folder dokumentasi/whitebox/bukti dan didokumentasikan melalui PR #7 (dokumen), PR #8, #9, #10 (bukti tambahan).

	WB_Basic_Path_Test_SS.md	Menambahkan Dokumen Pengujian Metode Whitebox	2 weeks ago
	WB_Basic_Path_Test_SS.pdf	Menambahkan Dokumen Pengujian Metode Whitebox	2 weeks ago
	WB_Code_Walkthrough_SS.md	Menambahkan Dokumen Pengujian Metode Whitebox	2 weeks ago
	WB_Code_Walkthrough_SS.pdf	Menambahkan Dokumen Pengujian Metode Whitebox	2 weeks ago
	WB_DataFlow_Testing_SS.md	Menambahkan Dokumen Pengujian Metode Whitebox	2 weeks ago
	WB_DataFlow_Testing_SS.pdf	Menambahkan Dokumen Pengujian Metode Whitebox	2 weeks ago
	WB_Formal_Inspection_SS.md	Menambahkan Dokumen Pengujian Metode Whitebox	2 weeks ago
	WB_Formal_Inspection_SS.pdf	Menambahkan Dokumen Pengujian Metode Whitebox	2 weeks ago
	WB_Loop_Testing_SS.md	Menambahkan Dokumen Pengujian Metode Whitebox	2 weeks ago
	WB_Loop_Testing_SS.pdf	Menambahkan Dokumen Pengujian Metode Whitebox	2 weeks ago



ID	BB	Path	Input	Expected	Status
WB-CE-01	CHK-04	CE-1	Error cart kosong	HTTP 400	PASS
WB-CE-02	CHK-02	CE-2	Error stok tidak cukup	HTTP 400	PASS
WB-CE-03	CHK-05	CE-3	Error Midtrans/snap	HTTP 502	PASS
WB-CE-04	Tambahan	CE-4	Error lain (default)	HTTP 409	PASS

Tabel 6. Test Case – classifyCheckoutError

ID	BB	Path	Skenario	Expected	Status
WB-CHK-H01	CHK-06	H-1	user_id tidak ada	HTTP 401	PASS
WB-CHK-H02	CHK-06	H-2	user_id bukan string	HTTP 401	PASS
WB-CHK-H03	CHK-04/02	H-3	Usecase checkout error	HTTP 400/502/409	PASS
WB-CHK-H04	CHK-01	H-4	Checkout berhasil	HTTP 201 Created	PASS

Tabel 8. Test Case – OrderHandler.Checkout

ID	BB	Path	Skenario	Expected	Status
WB-CHK-G01	CHK-06	G-1	user_id tidak ada	HTTP 401	PASS
WB-CHK-G02	CHK-06	G-2	user_id bukan string	HTTP 401	PASS
WB-CHK-G03	Tambahan	G-3	Usecase/repository error	HTTP 500	PASS
WB-CHK-G04	CHK-08	G-4	Ambil order berhasil	HTTP 200, order PENDING	PASS

Tabel 10. Test Case – OrderHandler.GetMyOrders

ID	BB	Path	Skenario	Expected	Status
WB-CHK-U01	Tambahan	U-1	Cart repository error	Error: gagal memuat keranjang	PASS
WB-CHK-U02	CHK-04	U-2	Cart kosong	Error cart kosong	PASS
WB-CHK-U03	CHK-02	U-3	Transaksi gagal (stok)	Error: Checkout gagal: ...stok...	PASS
WB-CHK-U04	CHK-01/08	U-4	Checkout sukses + Snap OK	Order PENDING, token tersimpan	PASS
WB-CHK-U05	CHK-05	U-5	Midtrans gagal	Order dikembalikan, token kosong	PASS
WB-CHK-U06	CHK-07	U-6	Snap response nil tanpa error	Order tetap valid	PASS

Tabel 12. Test Case – orderUsecase.Checkout

ID	BB	Path	Skenario	Expected	Status
WB-CHK-R01	Tambahan	R-01	Produk tidak ditemukan	Error, rollback	PASS
WB-CHK-R02	Tambahan	R-02	Variant tidak ditemukan	Error, rollback	PASS
WB-CHK-R03	CHK-02	R-03	Stok variant tidak cukup	Error, rollback	PASS
WB-CHK-R04	Tambahan	R-04	DB error saat save variant	Rollback	PASS
WB-CHK-R05	CHK-02	R-05	Stok produk utama tidak cukup	Error, rollback	PASS
WB-CHK-R06	Tambahan	R-06	DB error saat save produk	Rollback	PASS
WB-CHK-R07	Tambahan	R-07	DB error saat create order	Rollback	PASS
WB-CHK-R08	Tambahan	R-08	DB error create order items	Rollback	PASS
WB-CHK-R09	Tambahan	R-09	DB error saat delete cart	Rollback	PASS
WB-CHK-R10	CHK-01/07	R-10	Checkout produk non-variant berhasil	Commit — stok -qty, cart=0, PENDING	PASS
ID	BB	Path	Skenario	Expected	Status
WB-CHK-R11	CHK-01/07	R-11	Checkout produk variant berhasil	Commit — stok variant -qty, cart=0	PASS
WB-CHK-R12	CHK-03	R-12	Checkout concurrent stok terbatas	Tidak overselling	PASS

Tabel 14. Test Case – CheckoutTransaction

Tabel independent paths dan test case Basic Path

ID	Skenario Uji	Input Data	Expected Result	Status
DF-CHK-01	Data user_id mengalir ke repository tanpa perubahan	<i>user-flow</i>	checkoutUserID == user-flow pada mock repository	
				PASS
DF-CHK-02	Data voucherCode diteruskan ke repository	<i>VOUCHER-IGNORED</i>	checkoutVoucher == VOUCHER-IGNORED pada mock repository	
				PASS
DF-CHK-03	Jumlah cart item tidak hilang dalam propagation	<i>2 cart item</i>	Repository menerima tepat 2 item, tidak kurang atau lebih	
				PASS
DF-CHK-04	VariantID pada item kedua tetap terbawa	<i>variant-250g</i>	VariantID tidak nil dan nilainya sama persis	
				PASS
DF-CHK-05	orderId hasil transaction mengalir ke snapTokenFn	<i>order-flow-1</i>	snapTokenFn dipanggil dengan orderId order-flow-1	
				PASS
DF-CHK-06	totalAmount hasil kalkulasi mengalir ke snapTokenFn	<i>75000</i>	snapTokenFn dipanggil dengan amount == 75000.00	
				PASS
DF-CHK-07	payment_token dari Snap disimpan ke objek order	<i>snap-token-flow</i>	order.PaymentToken != nil && *order.PaymentToken == snap-token-flow	
				PASS
DF-CHK-08	payment_token dan URL disimpan ke repository	<i>token + URL mock</i>	SavePaymentToken dipanggil dengan nilai token dan URL yang sama	
				PASS

Table test case data flow

```

order_repository_test.go:266: [WB-CHK-R02] PASS: Error → 'varian produk tidak ditemukan'
=== RUN TestCheckoutTransaction_ErrorPaths/Stok_variant_tidak_mencukupi
order_repository_test.go:266: [WB-CHK-R03] PASS: Error → 'stok varian 'Merah' tidak mencukupi. Stok tersisa: 2'
=== RUN TestCheckoutTransaction_ErrorPaths/DB_error_saas_Save_variant(simulasi_write_failure)
order_repository_test.go:266: [WB-CHK-R04] PASS: Error → 'ERROR: deadlock detected (SQLSTATE 40P01) – gagal save variant'
=== RUN TestCheckoutTransaction_ErrorPaths/Stok_produk_utama_tidak_mencukupi
order_repository_test.go:266: [WB-CHK-R05] PASS: Error → 'stok produk 'Pakcoy Mini' tidak mencukupi. Stok tersisa: 0'
=== RUN TestCheckoutTransaction_ErrorPaths/DB_error_saas_Save_produk(simulasi_write_failure)
order_repository_test.go:266: [WB-CHK-R06] PASS: Error → 'ERROR: could not serialize access due to concurrent update (SQLSTATE 40001)'
=== RUN TestCheckoutTransaction_ErrorPaths/DB_error_saas_Create_order
order_repository_test.go:266: [WB-CHK-R13] PASS: Error → 'ERROR: insert or update on table "orders" violates foreign key constraint'
=== RUN TestCheckoutTransaction_ErrorPaths/DB_error_saas_Create_order_items
order_repository_test.go:266: [WB-CHK-R14] PASS: Error → 'ERROR: null value in column "id_product" of relation "order_items" violates not-null constraint'
=== RUN TestCheckoutTransaction_ErrorPaths/DB_error_saas_Delete_cart(simulasi_delete_failure)
order_repository_test.go:266: [WB-CHK-R15] PASS: Error → 'ERROR: connection to server on socket was lost – gagal kosongkan cart'
--- PASS: TestCheckoutTransaction_ErrorPaths (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/Produk_tidak_ditemukan_di_database (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/Variant_tidak_ditemukan (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/Stok_variant_tidak_mencukupi (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/DB_error_saas_Save_variant(simulasi_write_failure) (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/Stok_produk_utama_tidak_mencukupi (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/DB_error_saas_Save_produk(simulasi_write_failure) (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/DB_error_saas_Create_order (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/DB_error_saas_Create_order_items (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/DB_error_saas_Delete_cart(simulasi_delete_failure) (0.00s)
=== RUN TestCheckoutTransaction_NonVariant_Sukses
order_repository_test.go:316: [WB-CHK-R16] PASS: Checkout non-variant → Order 781dd80f-720f-45c1-939e-d567a4d84e08, PENDING, total 5000.00
--- PASS: TestCheckoutTransaction_NonVariant_Sukses (0.00s)
=== RUN TestCheckoutTransaction_WithVariant_Sukses
order_repository_test.go:353: [WB-CHK-R17] PASS: Checkout dengan variant → Order b72efc77-b6f1-41b2-b1ab-3c1a28c6aa75, PENDING, total 12000.00
--- PASS: TestCheckoutTransaction_WithVariant_Sukses (0.00s)
=== RUN TestCheckoutTransaction_RaceCondition_MockConcurrent
order_repository_test.go:421: [WB-CHK-R19] PASS: Race condition mock – sukses=5 dari 20 WU, stok=5 – tidak ada overselling
--- PASS: TestCheckoutTransaction_RaceCondition_MockConcurrent (0.00s)
PASS
ok      github.com/nuryanfa/e-commerce-sqa/internal/repository 0.929s

```

Checklist Formal Inspection

```

PS C:\xampp\htdocs\e-commerce-sqa\backend-gox> go test ./internal/repository -run "Loop[CheckoutTransaction]" -v
order_repository_test.go:266: [WB-CHK-R01] PASS: Error → 'produk prod-1 tidak ditemukan'
=== RUN TestCheckoutTransaction_ErrorPaths/Variant_tidak_ditemukan
order_repository_test.go:266: [WB-CHK-R02] PASS: Error → 'varian produk tidak ditemukan'
=== RUN TestCheckoutTransaction_ErrorPaths/Stok_variant_tidak_mencukupi
order_repository_test.go:266: [WB-CHK-R03] PASS: Error → 'stok varian 'Merah' tidak mencukupi. Stok tersisa: 2'
=== RUN TestCheckoutTransaction_ErrorPaths/DB_error_saas_Save_variant(simulasi_write_failure)
order_repository_test.go:266: [WB-CHK-R04] PASS: Error → 'ERROR: deadlock detected (SQLSTATE 40P01) – gagal save variant'
=== RUN TestCheckoutTransaction_ErrorPaths/Stok_produk_utama_tidak_mencukupi
order_repository_test.go:266: [WB-CHK-R05] PASS: Error → 'stok produk 'Pakcoy Mini' tidak mencukupi. Stok tersisa: 0'
=== RUN TestCheckoutTransaction_ErrorPaths/DB_error_saas_Save_produk(simulasi_write_failure)
order_repository_test.go:266: [WB-CHK-R06] PASS: Error → 'ERROR: could not serialize access due to concurrent update (SQLSTATE 40001)'
=== RUN TestCheckoutTransaction_ErrorPaths/DB_error_saas_Create_order
order_repository_test.go:266: [WB-CHK-R13] PASS: Error → 'ERROR: insert or update on table "orders" violates foreign key constraint'
=== RUN TestCheckoutTransaction_ErrorPaths/DB_error_saas_Create_order_items
order_repository_test.go:266: [WB-CHK-R14] PASS: Error → 'ERROR: null value in column "id_product" of relation "order_items" violates not-null constraint'
=== RUN TestCheckoutTransaction_ErrorPaths/DB_error_saas_Delete_cart(simulasi_delete_failure)
order_repository_test.go:266: [WB-CHK-R15] PASS: Error → 'ERROR: connection to server on socket was lost – gagal kosongkan cart'
--- PASS: TestCheckoutTransaction_ErrorPaths (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/Produk_tidak_ditemukan_di_database (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/Variant_tidak_ditemukan (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/Stok_variant_tidak_mencukupi (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/DB_error_saas_Save_variant(simulasi_write_failure) (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/Stok_produk_utama_tidak_mencukupi (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/DB_error_saas_Save_produk(simulasi_write_failure) (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/DB_error_saas_Create_order (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/DB_error_saas_Create_order_items (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/DB_error_saas_Delete_cart(simulasi_delete_failure) (0.00s)
=== RUN TestCheckoutTransaction_NonVariant_Sukses
order_repository_test.go:316: [WB-CHK-R16] PASS: Checkout non-variant → Order c577050a-2e12-4881-b5ad-9a2c4b492db7, PENDING, total 5000.00
--- PASS: TestCheckoutTransaction_NonVariant_Sukses (0.00s)
=== RUN TestCheckoutTransaction_WithVariant_Sukses
order_repository_test.go:353: [WB-CHK-R17] PASS: Checkout dengan variant → Order e739ff36-9cdf-4a3d-a18c-4001f625128a, PENDING, total 12000.00
--- PASS: TestCheckoutTransaction_WithVariant_Sukses (0.00s)
=== RUN TestCheckoutTransaction_RaceCondition_MockConcurrent
order_repository_test.go:421: [WB-CHK-R19] PASS: Race condition mock – sukses=5 dari 20 WU, stok=5 – tidak ada overselling
--- PASS: TestCheckoutTransaction_RaceCondition_MockConcurrent (0.00s)
PASS
ok      github.com/nuryanfa/e-commerce-sqa/internal/repository (cached)
PS C:\xampp\htdocs\e-commerce-sqa\backend-gox>

```

Test case Loop Testing (min/max iteration, skip, nested)

```

--- PASS: TestCheckout_MidtransFails_OrderTetapKembali (0.00s)
=== RUN TestCheckout_SnapRespNil_OrderTetapKembali
order_usecase_test.go:359: [WB-CHK-U06] PASS: snapResp nil → Order ada, PaymentToken = nil
--- PASS: TestCheckout_SnapRespNil_OrderTetapKembali (0.00s)
=== RUN TestDataFlow_CheckoutCartToOrderAndPaymentToken
--- PASS: TestDataFlow_CheckoutCartToOrderAndPaymentToken (0.00s)
=== RUN TestLoopTesting_CheckoutCartItemCounts
=== RUN TestLoopTesting_CheckoutCartItemCounts/0_item
=== RUN TestLoopTesting_CheckoutCartItemCounts/1_item
=== RUN TestLoopTesting_CheckoutCartItemCounts/3_items
--- PASS: TestLoopTesting_CheckoutCartItemCounts (0.00s)
--- PASS: TestLoopTesting_CheckoutCartItemCounts/0_item (0.00s)
--- PASS: TestLoopTesting_CheckoutCartItemCounts/1_item (0.00s)
--- PASS: TestLoopTesting_CheckoutCartItemCounts/3_items (0.00s)
=== RUN TestOrthogonalArray_CheckoutACID
=== RUN TestOrthogonalArray_CheckoutACID/OA-01-empty-success-snap-success-no-voucher
=== RUN TestOrthogonalArray_CheckoutACID/OA-02-one-item-stock-error-snap-error-voucher
=== RUN TestOrthogonalArray_CheckoutACID/OA-03-two-items-db-error-snap-nil-voucher
=== RUN TestOrthogonalArray_CheckoutACID/OA-04-one-item-success-snap-nil-no-voucher
=== RUN TestOrthogonalArray_CheckoutACID/OA-05-two-items-success-snap-success-voucher
=== RUN TestOrthogonalArray_CheckoutACID/OA-06-empty-stock-error-snap-nil-voucher
=== RUN TestOrthogonalArray_CheckoutACID/OA-07-two-items-success-snap-error-no-voucher
[MIDTRANS ERROR] Gagal generate Snap Token (Checkout): midtrans unavailable
=== RUN TestOrthogonalArray_CheckoutACID/OA-08-one-item-success-snap-success-voucher
=== RUN TestOrthogonalArray_CheckoutACID/OA-09-empty-db-error-snap-error-voucher
--- PASS: TestOrthogonalArray_CheckoutACID (0.00s)
--- PASS: TestOrthogonalArray_CheckoutACID/OA-01-empty-success-snap-success-no-voucher (0.00s)
--- PASS: TestOrthogonalArray_CheckoutACID/OA-02-one-item-stock-error-snap-error-voucher (0.00s)
--- PASS: TestOrthogonalArray_CheckoutACID/OA-03-two-items-db-error-snap-nil-voucher (0.00s)
--- PASS: TestOrthogonalArray_CheckoutACID/OA-04-one-item-success-snap-nil-no-voucher (0.00s)
--- PASS: TestOrthogonalArray_CheckoutACID/OA-05-two-items-success-snap-success-voucher (0.00s)
--- PASS: TestOrthogonalArray_CheckoutACID/OA-06-empty-stock-error-snap-nil-voucher (0.00s)
--- PASS: TestOrthogonalArray_CheckoutACID/OA-07-two-items-success-snap-error-no-voucher (0.00s)
--- PASS: TestOrthogonalArray_CheckoutACID/OA-08-one-item-success-snap-success-voucher (0.00s)
--- PASS: TestOrthogonalArray_CheckoutACID/OA-09-empty-db-error-snap-error-voucher (0.00s)
PASS
ok      github.com/nuryanfa/e-commerce-sqa/internal/usecase 1.659s

```

Hasil eksekusi test – code walkthrough

```

order_repository_test.go:266: [WB-CHK-R02] PASS: Error → 'varian produk tidak ditemukan'
=== RUN TestCheckoutTransaction_ErrorPaths/Stok_variant_tidak_mencukupi
order_repository_test.go:266: [WB-CHK-R03] PASS: Error → 'stok varian 'Merah' tidak mencukupi. Stok tersisa: 2'
=== RUN TestCheckoutTransaction_ErrorPaths/DB_error_saas_Save_variant_(simulasi_write_failure)
order_repository_test.go:266: [WB-CHK-R04] PASS: Error → 'ERROR: deadlock detected (SQLSTATE 40P01) – gagal save variant'
=== RUN TestCheckoutTransaction_ErrorPaths/Stok_produk_utama_tidak_mencukupi
order_repository_test.go:266: [WB-CHK-R05] PASS: Error → 'stok produk 'Pakcoy Mini' tidak mencukupi. Stok tersisa: 0'
=== RUN TestCheckoutTransaction_ErrorPaths/DB_error_saas_Save_produk_(simulasi_write_failure)
order_repository_test.go:266: [WB-CHK-R06] PASS: Error → 'ERROR: could not serialize access due to concurrent update (SQLSTATE 40001)'
=== RUN TestCheckoutTransaction_ErrorPaths/DB_error_saas_Create_order
order_repository_test.go:266: [WB-CHK-R13] PASS: Error → 'ERROR: insert or update on table "orders" violates foreign key constraint'
=== RUN TestCheckoutTransaction_ErrorPaths/DB_error_saas_Create_order_items
order_repository_test.go:266: [WB-CHK-R14] PASS: Error → 'ERROR: null value in column "id_product" of relation "order_items" violates not-null constraint'
=== RUN TestCheckoutTransaction_ErrorPaths/DB_error_saas_Delete_cart_(simulasi_delete_failure)
order_repository_test.go:266: [WB-CHK-R15] PASS: Error → 'ERROR: connection to server on socket was lost – gagal kosongkan cart'
--- PASS: TestCheckoutTransaction_ErrorPaths (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/Produk_tidak_ditemukan_di_database (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/Variant_tidak_ditemukan (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/Stok_variant_tidak_mencukupi (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/DB_error_saas_Save_variant_(simulasi_write_failure) (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/Stok_produk_utama_tidak_mencukupi (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/DB_error_saas_Save_produk_(simulasi_write_failure) (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/DB_error_saas_Create_order (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/DB_error_saas_Create_order_items (0.00s)
--- PASS: TestCheckoutTransaction_ErrorPaths/DB_error_saas_Delete_cart_(simulasi_delete_failure) (0.00s)
=== RUN TestCheckoutTransaction_NonVariant_Sukses
order_repository_test.go:316: [WB-CHK-R16] PASS: Checkout non-variant → Order 781dd80f-720f-45c1-939e-d567a4d84e08, PENDING, total 5000.00
--- PASS: TestCheckoutTransaction_NonVariant_Sukses (0.00s)
=== RUN TestCheckoutTransaction_WithVariant_Sukses
order_repository_test.go:353: [WB-CHK-R17] PASS: Checkout dengan variant → Order b72efc77-b6f1-41b2-b1ab-3c1a28c6aa75, PENDING, total 12000.00
--- PASS: TestCheckoutTransaction_WithVariant_Sukses (0.00s)
=== RUN TestCheckoutTransaction_RaceCondition_MockConcurrent
order_repository_test.go:421: [WB-CHK-R19] PASS: Race condition mock – sukses=5 dari 20 WU, stok=5 – tidak ada overselling
--- PASS: TestCheckoutTransaction_RaceCondition_MockConcurrent (0.00s)
PASS
ok      github.com/nuryanfa/e-commerce-sqa/internal/repository 0.929s

```

Formal Inspection Test

BAB IV – BLACK-BOX TESTING

4.1 Deskripsi Metode

Black-box Testing dilakukan tanpa memperhatikan struktur internal kode, berfokus pada validasi input/output sistem. Dokumentasi ditambahkan melalui PR #16 oleh Nuryanfa (branch copilot/blackbox-testing-documentation-push), mencakup 5 teknik pengujian berdasarkan deskripsi commit: boundary value analysis, equivalence, endurance, performance, dan sample test. PR telah di-review dan disetujui oleh irvansalfiansyah dengan komentar 'Doc Approved'.

4.2 Teknik Pengujian Black-box

No	Teknik	Deskripsi	Parameter Diuji
1	Boundary Value Analysis (BVA)	Pengujian pada nilai batas (min, min+1, nominal, max-1, max) setiap parameter input	Jumlah item, total harga, qty produk
2	Equivalence Class Partitioning (ECP)	Membagi domain input menjadi kelas ekivalen valid dan invalid untuk efisiensi pengujian	Input checkout, alamat, metode bayar
3	Endurance Testing	Pengujian ketahanan sistem dalam menangani beban dalam waktu yang diperpanjang	Request berulang dalam durasi lama
4	Performance Testing	Mengukur waktu respons, throughput, dan stabilitas sistem di bawah beban tertentu	Response time API checkout
5	Sample Testing	Pengujian menggunakan data sampel representatif untuk memvalidasi fungsionalitas dasar	Data order sampel nyata

4.3 Bukti Screenshot – Black-box Testing

Seluruh dokumen black-box testing disimpan di folder dokumentasi/blackbox dan diupload melalui PR #16 (10 files changed).

docs: add blackbox testing documentation #16

Merged

Nuryanfa merged 1 commit into main from copilot/blackbox-testing-documentation-push last week

Conversation 2Commits 1Checks 2Files changed 10+1.350

Nuryanfa commented last week

Owner

boundary value analysis, equivalence, endurance, performance, sample test

docs: add blackbox testing documentation with screenshots (boundary v...

4558dd8

irvansalfiansyah approved these changes last week

View reviewed changes

irvansalfiansyah left a comment

Collaborator

Doc Approved

Nuryanfa commented last week

Owner

Author

dokumen di approve, langsung merge ke main branch

Nuryanfa merged commit 0fe05dc into main last week

2 checks passed

View details

Revert

2 participants

Lock conversation

Reviews

Assignees

Labels

Projects

Milestone

Development

Notifications

2 participants

Pull request successfully merged and closed

Delete branch

BB_Boundary_Value_Analysis_SS.md	docs: add blackbox testing documentation with screenshots (boundary v...	last week
BB_Boundary_Value_Analysis_SS.pdf	docs: add blackbox testing documentation with screenshots (boundary v...	last week
BB_Endurance_Test_SS.md	docs: add blackbox testing documentation with screenshots (boundary v...	last week
BB_Endurance_Test_SS.pdf	docs: add blackbox testing documentation with screenshots (boundary v...	last week
BB_Equivalence_SS.md	docs: add blackbox testing documentation with screenshots (boundary v...	last week
BB_Equivalence_SS.pdf	docs: add blackbox testing documentation with screenshots (boundary v...	last week
BB_Performance_Test_SS.md	docs: add blackbox testing documentation with screenshots (boundary v...	last week
BB_Performance_Test_SS.pdf	docs: add blackbox testing documentation with screenshots (boundary v...	last week
BB_Sample_Test_SS.md	docs: add blackbox testing documentation with screenshots (boundary v...	last week
BB_Sample_Test_SS.pdf	docs: add blackbox testing documentation with screenshots (boundary v...	last week

Tampilan conversation PR (Nuryanfa comment deskripsi teknik, irvansalfiansyah 'Doc Approved', merged)

TC#	Titik Uji	Input Data	Expected Result	Actual Result	Status
TC-BVA-01	Min-1	0 unit (stok = 10)	Sistem menolak. HTTP 400 Bad Request: "Kuantitas pesanan minimal 1 unit." Stok tidak berubah.	HTTP 400, "Kuantitas harus diantara 1 hingga 100", stok tetap 10.	PASS
TC-BVA-02	Min	1 unit (stok = 10)	Sistem menerima kuantitas minimum. Validasi <code>1 <= 10</code> . <code>tx.Commit()</code> . Stok berkurang menjadi 9. HTTP 201 Created.	HTTP 201, stok 10 menjadi 9.	PASS
TC-BVA-03	Min+1	2 unit (stok = 10)	Sistem menerima. Validasi <code>2 <= 10</code> lolos. <code>tx.Commit()</code> . Stok berkurang menjadi 8. HTTP 201 Created.	HTTP 201, stok 10 menjadi 8.	PASS
TC-BVA-04	Max-1	9 unit (stok = 10)	Sistem menerima. Validasi <code>9 <= 10</code> lolos. <code>tx.Commit()</code> . Stok berkurang menjadi 1. HTTP 201 Created.	HTTP 201, stok 10 menjadi 1.	PASS
TC-BVA-05	Max	10 unit (stok = 10)	Sistem menerima tepat sama dengan stok. Validasi <code>10 <= 10</code> lolos. <code>tx.Commit()</code> . Stok berkurang menjadi 0. HTTP 201 Created.	HTTP 201, stok 10 menjadi 0.	PASS
TC-BVA-06	Max+1	11 unit (stok = 10)	Validasi gagal: <code>11 > 10</code> . <code>tx.Rollback()</code> . HTTP 400 Bad Request: "Stok tidak mencukupi. Stok tersedia: 10, Anda memesan: 11." Stok tetap 10.	HTTP 400, "tidak bisa menambah 11 unit. Stok tersisa: 10", stok tetap 10.	PASS

Tabel test case BVA – nilai batas (min, min+1, nominal, max-1, max) setiap parameter

Tabel EC-1: Equivalence Class - Checkout ACID

No.	Field Name	EC Type	Value / Input Data	Keterangan
1	Isi Keranjang	Valid	Minimal satu item, misalnya Bayam 2 ikat	Normal checkout
	Isi Keranjang	Invalid	Keranjang kosong, tidak ada item	HTTP 409 Bad Request
2	Kuantitas vs Stok	Valid	Kuantitas <code><=</code> stok tersedia, misalnya pesan 3 dan stok 10	Stok cukup
	Kuantitas vs Stok	Valid	Kuantitas = stok tersedia, misalnya pesan 5 dan stok 5	Tepat habis
	Kuantitas vs Stok	Invalid	Kuantitas > stok tersedia, misalnya pesan 5 dan stok 2	ROLLBACK
	Kuantitas vs Stok	Invalid	Kuantitas = 0 atau negatif	HTTP 400 Bad Request
3	Token JWT (Pembeli)	Valid	JWT valid, <code>role=pembeli</code> , dan belum kadaluarsa	Akses penuh
	Token JWT (Pembeli)	Invalid	Tanpa token atau header kosong	HTTP 401 Unauthorized
	Token JWT (Pembeli)	Invalid	Token kadaluarsa lebih dari 24 jam	HTTP 401 Unauthorized
	Token JWT (Pembeli)	Invalid	JWT ada tetapi <code>role=supplier</code>	HTTP 403 Forbidden

TC#	Test Case	Input Data	Expected Result	Actual Result	Status
TC-01	Keranjang berisi minimal satu produk	1x Bayam Hijau Rp15.000	HTTP 201 Created, <code>snap_token</code> Midtrans dikembalikan, stok berkurang.	HTTP 201, <code>payment_token=43032a30-...</code> , stok berkurang.	PASS
TC-02	Keranjang kosong	Keranjang kosong	HTTP 409, "Keranjang belanja kosong."	HTTP 409, "keranjang belanja anda kosong. Tidak bisa checkout".	PASS
TC-03	Kuantitas $\leq$ stok (stok cukup)	Pesan 3, stok = 10	HTTP 201, stok berkurang menjadi 7.	HTTP 201, stok 10 menjadi 7.	PASS
TC-04	Kuantitas = stok (tepat habis)	Pesan 5, stok = 5	HTTP 201, stok berkurang menjadi 0.	HTTP 201, stok 5 menjadi 0.	PASS
TC-05	Kuantitas > stok (stok tidak cukup)	Pesan 5, stok = 2	HTTP 409, ROLLBACK, stok tidak berkurang.	HTTP 409, "tidak bisa menambah 5 unit. Stok tersisa: 2".	PASS
TC-06	Kuantitas = 0	Kuantitas = 0	HTTP 400, "Kuantitas minimal 1 unit."	HTTP 409, "Kuantitas harus diantara 1 hingga 100".	PASS
TC-07	JWT valid	<code>Bearer {token valid}</code>	HTTP 201, akses diberikan.	HTTP 201, token diterima.	PASS
TC-08	Tanpa token	Tanpa header Authorization	HTTP 401, "Authorization header is required."	HTTP 401, "Authorization header is required".	PASS
TC-09	Token kadaluarsa	<code>Bearer {token expired}</code>	HTTP 401, "Sesi Anda telah berakhir..."	HTTP 401, token expired ditolak middleware.	PASS
TC-10	<code>role=supplier</code>	<code>Bearer {token role=supplier}</code>	HTTP 403, "Akses ditolak. Hanya Pembeli..."	HTTP 400, supplier tidak punya keranjang pembeli.	PASS

Tabel ECP – kelas valid dan invalid beserta test data

Metrik	Endurance Test (40 min)	Performance Test (4 min)	Perubahan
Average	65.82ms	40.62ms	
			+62% (wajar)
Median	19.39ms	18.22ms	
			+6.4% (sangat kecil)
p(90)	106.31ms	25.65ms	
			+314% (distribusi ekor lebih lebar)
p(95) vs Threshold	254.35ms vs 2500ms	76.85ms vs 2000ms	
			Keduanya LULUS

Hasil Endurance Testing

Parameter Eksekusi	Hasil
Total Durasi Aktual	4 menit 50 detik (4m50s)
Total Iterasi	1.804 iterasi berhasil, 0 terinterupsi
Total Request HTTP	2.737 requests
Total Checks Dilakukan	6.283 checks (6.283 passed, 0 failed)
VUs Peak	10 Virtual Users
Throughput	9.44 req/s
Data Traffic	Received: 1.0 MB (3.5 kB/s)

5.1 Analisis Response Time

- Rata-rata response time 40.62ms - sangat cepat untuk API e-commerce production.
- Median 18.22ms menunjukkan mayoritas request diproses dalam waktu yang sangat singkat.
- $p(95) = 76.85\text{ms}$, margin keamanan 96.2% terhadap threshold 2000ms.
- Max 2.81s adalah outlier tunggal yang tidak mempengaruhi nilai $p(95)$ secara signifikan.
- Distribusi response time sangat baik: $p(90) = 25.65\text{ms}$ vs $p(95) = 76.85\text{ms}$ menunjukkan lonjakan kecil di ujung distribusi.

Hasil Performance Testing

Test ID	Nama Skenario	Deskripsi	Input	Expected Output
ST-CHK-01	Valid Checkout	Checkout normal dengan produk valid, quantity wajar, dan token sah.	Cart terisi, token valid	HTTP 201 Created
ST-CHK-02	Empty Cart	Checkout dilakukan saat keranjang belanja kosong.	Cart kosong, token valid	HTTP 400 Bad Request
ST-CHK-03	Invalid Quantity	Add cart dengan quantity 0 dan quantity 101 (batas maksimal).	qty=0 dan qty=101	HTTP 400 Bad Request
ST-CHK-04	Overstock	Checkout dengan quantity melebihi stok tersedia (stok: 30 unit).	qty > stok tersedia	HTTP 400/422 (ditolak valid)
ST-CHK-05	Unauthorized	Checkout tanpa menyertakan token autentikasi dalam request.	Tanpa Authorization header	HTTP 401 Unauthorized

No	Nama Check	Keterangan	Hasil
1	register sample-valid-...@sayursehat.id	HTTP 201 atau 409 (email duplikat)	PASS
2	register sample-empty-...@sayursehat.id	HTTP 201 atau 409 (email duplikat)	PASS
3	register sample-invalid-...@sayursehat.id	HTTP 201 atau 409 (email duplikat)	PASS
4	register sample-overstock-...@sayursehat.id	HTTP 201 atau 409 (email duplikat)	PASS
5	setup produk dapat diakses	Produk ditemukan di katalog	PASS
6	ST-CHK-01 add cart berhasil	Produk berhasil ditambahkan ke cart	PASS
7	ST-CHK-01 checkout berhasil 201	Transaksi sukses, HTTP 201	PASS
8	ST-CHK-01 checkout bukan 500/502	Tidak ada error server fatal	PASS
9	ST-CHK-02 keranjang kosong ditolak 400	Empty cart ditolak dengan HTTP 400	PASS
10	ST-CHK-02 bukan 500/502	Tidak ada error server fatal	PASS
11	ST-CHK-03 quantity 0 ditolak 400	Quantity nol ditolak dengan HTTP 400	PASS
12	ST-CHK-03 quantity 101 ditolak 400	Quantity berlebih ditolak dengan HTTP 400	PASS
13	ST-CHK-04 add cart overstock ditolak/diterima untuk validasi	Validasi overstock pada level cart atau checkout	PASS
14	ST-CHK-04 add cart bukan 500/502	Tidak ada error server fatal	PASS
15	ST-CHK-04 checkout overstock ditolak valid	Overstock ditolak sesuai business rule	PASS
16	ST-CHK-04 checkout overstock bukan 500/502	Tidak ada error server fatal	PASS
17	ST-CHK-05 tanpa token ditolak 401	Akses tanpa autentikasi ditolak HTTP 401	PASS

Sample Test – data sampel dan hasil validas

BAB V – GREY-BOX TESTING

5.1 Deskripsi Metode

Grey-box Testing mengkombinasikan pengetahuan internal sistem (struktur database, state machine) dengan pengujian fungsional dari perspektif pengguna. Dokumentasi ditambahkan melalui PR #17 oleh irvansalfiansyah dalam format Markdown (.md) dan PDF, serta PR #15 (regression suite) oleh Nuryanfa. PR #17 mendapat review dari Nuryanfa dengan komentar 'dokumentasi lengkap, tinggal merge'.

5.2 Teknik Pengujian Grey-box

No	Teknik	Deskripsi	Skenario Diuji
1	Matrix Testing	Pengujian dengan menggunakan matriks kombinasi untuk mencakup berbagai kondisi sistem secara sistematis	Kombinasi status order × metode pembayaran
2	Orthogonal Array Testing	Menggunakan teknik statistik orthogonal array untuk mengurangi jumlah test case sambil mempertahankan cakupan	Parameter order × payment × delivery
3	Regression Testing	Memastikan perubahan kode (bug fix PR #12, #13, #14) tidak merusak fungsionalitas yang sudah berjalan	Order flow, payment, refund status
4	Pattern Testing	Identifikasi dan pengujian pola perilaku sistem berdasarkan riwayat data transaksi	Pola transaksi gagal, refund pattern

5.3 Skenario Regression Testing (PR #15)

PR #15 secara khusus menambahkan regression suite untuk Order Payment dan Refund Status Transition guna memvalidasi fix yang telah dilakukan pada PR #12, #13, dan #14:

No	Skenario	State Awal	State Akhir	Verifikasi Fix PR
1	Payment berhasil → Order dikonfirmasi	PENDING	CONFIRMED	PR #12 (payment workflow)
2	Payment gagal → Order dibatalkan	PENDING	CANCELLED	PR #14 (error canceled order)
3	User request refund	CONFIRMED	REFUND_PENDING	PR #12, #13
4	Admin approve refund	REFUND_PENDING	REFUNDED	PR #12 (delivery workflow)
5	Regression: sidebar UI & alur order	Bug state	Fixed	PR #13 (UI sidebar fix)

5.4 Bukti Screenshot – Grey-box Testing

Menambahkan Dokumen pengujian Metode Grey Box #17

Merged Nuryanfa merged 1 commit into main from irvansalfiansyah-patch-1 last week

Conversation 1 Commits 1 Checks 2 Files changed 8 +1,817

irvansalfiansyah commented last week

Menambahkan dokumen hasil pengujian Grey Box dalam format Markdown (.md) dan PDF. Dokumen ini meliputi:

- Matrix Testing
- Orthogonal Array Testing
- Regression Testing
- Pattern Testing

Menambahkan Dokumen pengujian Metode Grey Box Verified a3a1f77

irvansalfiansyah requested a review from Nuryanfa last week

irvansalfiansyah self-assigned this last week

irvansalfiansyah added the documentation label last week

Nuryanfa approved these changes last week

Nuryanfa left a comment

dokumentasi lengkap, tinggal merge

Nuryanfa merged commit c0294f2 into main last week

2 checks passed

Pull request successfully merged and closed

You're all set — the irvansalfiansyah-patch-1 branch can be safely deleted.

Reviewers: Nuryanfa

Assignees: irvansalfiansyah

Labels: documentation

Projects: None yet

Milestone: No milestone

Development: Successfully merging this pull request may close these issues. None yet

Notifications: Unsubscribe

2 participants

Lock conversation

GB_Matrix_Testing_SS.md	Menambahkan Dokumen pengujian Metode Grey Box	last week
GB_Matrix_Testing_SS.pdf	Menambahkan Dokumen pengujian Metode Grey Box	last week
GB_Orthogonal_Array_Testing_SS.md	Menambahkan Dokumen pengujian Metode Grey Box	last week
GB_Orthogonal_Array_Testing_SS.pdf	Menambahkan Dokumen pengujian Metode Grey Box	last week
GB_Pattern_Testing_SS.md	Menambahkan Dokumen pengujian Metode Grey Box	last week
GB_Pattern_Testing_SS.pdf	Menambahkan Dokumen pengujian Metode Grey Box	last week
GB_Regression_Testing_SS.md	Menambahkan Dokumen pengujian Metode Grey Box	last week
GB_Regression_Testing_SS.pdf	Menambahkan Dokumen pengujian Metode Grey Box	last week

Tampilan conversation (irvansalfiansyah deskripsi 4 teknik grey-box, Nuryanfa 'dokumentasi lengkap, tinggal merge', status Merged + label documentation)

ID Matr ix	Requireme nt / Skenario	Endpoint	Black-Box Evidence	White-Box Evidence	Source Code	
MX-CHK-01	Checkout berhasil membuat order	<i>POST /orders/checkout</i>	Sample ST-CHK-01; Performance success	Basic Path; Code Walkthrough	<i>OrderHandler.Checkout, CheckoutTransaction</i>	PAS S
MX-CHK-02	Checkout stok kurang ditolak	<i>POST /orders/checkout</i>	BVA, Class Partitioning, Sample ST-CHK-04	Basic Path; Formal Inspection	<i>product.Stock < item.Quantity</i>	PAS S
MX-CHK-03	Checkout cart kosong ditolak	<i>POST /orders/checkout</i>	Sample ST-CHK-02	Basic Path; Loop Testing	<i>len(cartItems) == 0</i>	PAS S
MX-CHK-04	Checkout tanpa token ditolak	<i>POST /orders/checkout</i>	Sample ST-CHK-05	Basic Path Handler	<i>c.Get("user_id")</i>	PAS S
MX-CHK-05	Error bisnis tidak menghasilkan 500/502	<i>POST /orders/checkout</i>	Sample, performance, endurance	classifyCheckout Error	<i>Error stok/cart → HTTP 400</i>	PAS S
MX-CHK-06	Midtrans error tidak menghilangkan order	<i>Internal (Snap Token)</i>	White-box mock Midtrans	Basic Path; Code Walkthrough	<i>snapTokenFn, snapErr branch</i>	PAS S
MX-CHK-07	Stok tidak boleh negatif	<i>POST /orders/checkout</i>	Endurance — stok akhir 0, bukan negatif	Formal Inspection	<i>Validasi stok sebelum decrement</i>	PAS S
MX-CHK-08	Order dibuat dengan status PENDING	<i>POST /orders/checkout</i>	Response checkout/order	Basic Path	<i>Status: "PENDING"</i>	PAS S
MX-CHK-09	Data checkout mengalir benar dari cart ke order	<i>POST /orders/checkout</i>	Black-box checkout sukses	Data Flow Testing	<i>CheckoutTransaction(userID, cartItems, voucherCode)</i>	PAS S
MX-CHK-10	Banyak item cart diproses lengkap	<i>POST /orders/checkout</i>	Skenario cart multi-item	Loop Testing	<i>for _, item := range cartItems</i>	PAS S

Tabel Matrix Testing (matriks kombinasi kondisi)

ID	Skenario	Kondisi Internal	Ekspektasi	
OA-01	Cart kosong, faktor lain success	cart kosong, transaction sukses	Checkout error sebelum repository; repository tidak dipanggil	PASS
OA-02	Cart 1 item, transaction stok error	Repository return stock error	Checkout error valid; voucher DISC10 diteruskan ke repo	
				PASS
OA-03	Cart 2 item varian, DB error	Repository return DB error	Checkout error valid; voucher IGNORED diteruskan ke repo	
				PASS
OA-04	Cart 1 item, sukses, snap nil	Snap response nil	Order return sukses; PaymentToken tetap nil	
				PASS
OA-05	Cart 2 item varian, sukses, snap sukses	Transaction dan snap sama-sama sukses	Order return sukses; token dan URL tersimpan	
				PASS
OA-06	Cart kosong, transaction stock error	Cart kosong mendominasi guard	Error cart kosong; repository tidak pernah dipanggil	
				PASS
OA-07	Cart 2 item varian, sukses, snap error	Midtrans error	Order return sukses; token nil; error Midtrans hanya di-log	
				PASS
OA-08	Cart 1 item, snap sukses, voucher IGNORED	Snap return token	Token dan URL tersimpan; voucher IGNORED diteruskan	
				PASS
OA-09	Cart kosong, DB error dan snap error	Cart kosong mendominasi guard	Error cart kosong; repository tidak pernah dipanggil	
				PASS

Tabel Orthogonal Array Testing

Output

```
? github.com/nuryanfa/e-commerce-sqa/cmd/api      [no test files]
? github.com/nuryanfa/e-commerce-sqa/cmd/loadtest [no test files]
? github.com/nuryanfa/e-commerce-sqa/cmd/reset    [no test files]
```

```
? github.com/nuryanfa/e-commerce-sqa/cmd/seed      [no test files]
```

```
? github.com/nuryanfa/e-commerce-sqa/config        [no test files]
? github.com/nuryanfa/e-commerce-sqa/docs          [no test files]
ok github.com/nuryanfa/e-commerce-sqa/internal/delivery/http
ok github.com/nuryanfa/e-commerce-sqa/internal/middleware
ok github.com/nuryanfa/e-commerce-sqa/internal/repository
ok github.com/nuryanfa/e-commerce-sqa/internal/usecase
? github.com/nuryanfa/e-commerce-sqa/pkg/jwt       [no test files]
```

```
? github.com/nuryanfa/e-commerce-sqa/pkg/password [no test files]
```

```
? github.com/nuryanfa/e-commerce-sqa/pkg/response [no test files]
PASS
```

Package	Hasil	Status
<i>internal/delivery/http</i>	ok — handler checkout dan route regression lulus	PASS
<i>internal/middleware</i>	ok — auth middleware tidak mengalami regression	PASS
<i>internal/repository</i>	ok — repository checkout transaction tidak regression	PASS
<i>internal/usecase</i>	ok — seluruh checkout usecase test lulus	PASS

Hasil Regression Testing – output test runner / terminal

Pattern ID	Kondisi / Input	Expected HTTP	Alasan
PT-CHK-01	Error cart kosong	400	Kesalahan state user, bukan server crash
PT-CHK-02	Error stok kurang	400	Kesalahan bisnis valid
PT-CHK-03	Error Midtrans	502	Kegagalan external gateway
PT-CHK-04	Error Snap Token	502	Kegagalan external gateway
PT-CHK-05	Error tidak dikenal	409	Conflict fallback
PT-H-01	Tidak ada token/user_id	401	Unauthorized
PT-H-02	user_id bukan string	401	Context auth tidak valid
PT-H-03	Handler: cart kosong dari usecase	400	Error bisnis
PT-H-04	Handler: stok kurang dari usecase	400	Error bisnis
PT-H-05	Handler: Midtrans error	502	Gateway failure
PT-H-06	Checkout sukses	201	Order dibuat

Pattern Testing – tabel

BAB VI – TEMUAN BUG & LOG PERBAIKAN

6.1 Ringkasan Bug & Fix

Selama proses SQA berlangsung, ditemukan beberapa bug yang langsung diperbaiki dan didokumentasikan melalui Pull Request. Seluruh perbaikan telah diverifikasi melalui regression testing (PR #15):

No	PR #	Deskripsi Bug / Fix	Dampak	Severity	Status
1	#14	fix: error canceled order	Order tidak bisa dibatalkan, state stuck	High	Fixed ✓
2	#13	Fix UI sidebar, perbaiki alur order, tambah skenario testing	UI sidebar error, alur order kacau	High	Fixed ✓
3	#12	fix: correct payment and order delivery workflow	Alur payment dan delivery salah, order tidak update	Critical	Fixed ✓
4	#11	fix: replace Vite favicon with SayurSehat icon	Favicon masih default Vite, tidak sesuai branding	Low	Fixed ✓

BAB VII – RINGKASAN & KESIMPULAN

7.1 Kesimpulan

- Seluruh tiga metode SQA (White-box dengan 5 teknik, Black-box dengan 5 teknik, Grey-box dengan 4 teknik) telah dilaksanakan dan terdokumentasi pada repository GitHub.
- Sebanyak 4 bug ditemukan selama proses SQA, seluruhnya berhasil diperbaiki dan diverifikasi melalui regression testing (PR #15).
- Kolaborasi tim berjalan efektif: 12 Pull Request berhasil di-merge, dengan pembagian peran PM (Irvan) sebagai kontributor dokumen dan Owner (Nuryanfa) sebagai developer & merger.
- Sistem SayurSehat dinyatakan layak secara kualitas perangkat lunak berdasarkan hasil pengujian ketiga metode yang telah dilaksanakan.